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Solutions for optimizing heat dissipation on the printed circuit board of the drive circuit of a DC motor

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Abstract: H-bridge circuits are widely used in modern electronics to control DC (DC) motors. H-bridge circuits are power electronics systems consisting of MOSFETs or transistors to control the direction and speed of the motor. However, due to the high current carrying capacity and switching losses of the H-bridge circuit, heat dissipation poses a critical problem. Optimizing heat dissipation in PCB design is essential to increase the long-term reliability of the circuit and prevent heat buildup. In this article, the methods used for thermal optimization in PCB design of H-bridge circuit will be investigated. First, the heat distribution of the PCB will be analyzed using simulation methods. The goal is to determine the optimal design principles to improve the thermal efficiency of the PCB.

Keywords: H-bridge circuits, PCB, MOSFET, motors, modern electronics, optimization.

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1. Introduction

Proper heat dissipation in PCB design is crucial for the reliable operation of high-power motor drive systems. The existing scientific literature presents various optimization approaches in this area and recommends effective implementation methods in practice. For example, the article "Advanced Thermal Management Solutions on PCBs for High Power Applications" (2014) by M. Töpper, R. Aschenbrenner and H. Reichl notes that the efficiency of thermal management in PCBs designed for high-power applications can be significantly increased by using thick copper layers and thermal vias. The authors' analysis shows that these methods allow for a homogeneous temperature distribution on the PCB and the prevention of localized heating. . At the same time, the article "Thermal Design Optimization of Electronic Circuit Board Layout with Consideration of Natural Convection and Radiation" (2021) by Y. Zhang, X. Luo, and M. Liu emphasizes the importance of considering passive cooling mechanisms such as natural convection and thermal radiation in PCB design. According to the results of the article, proper placement of components and maintaining gaps for airflow can ensure the temperature stability of the circuit. On the other hand, the article "Thermal Modeling and Design Optimization of PCB Vias and Pads" (2018) by Y. Zhou, D. A. Staton, and C. M. Bingham presents analytical modeling and design optimization of vias and pads in PCB in terms of heat dissipation. Simulation results prove that the density of vias and their placement geometry directly affects the thermal resistance and the overall temperature behavior of the circuit. The PCB design was prepared taking into account these scientific recommendations: Power components were placed on large copper areas, Multiple thermal vias were added under each power transistor, Thermal gaps were maintained between components to allow unhindered airflow. Simulation results conducted in the first stage and then real tests conducted using a thermal camera showed that this approach was successful and confirmed that the temperature was distributed homogeneously over the circuit [1-10].

2. Experimental detail

The aim of this study was to optimize the heat dissipation in the PCB design of an H-bridge type driver circuit used to control DC motors. Practical design, combined with thermal simulation, was applied to balance the heat flow and prevent overheating of power components.

2.1. PCB Design Preparation

At the initial stage, the schematic structure of the H-bridge configuration power circuit was prepared and the placement of power components (especially MOSFETs) was planned. The PCB was designed in a 2-layer (bifacial) structure, where the top layer (Top Layer) was mainly used for signal and power paths, and the bottom layer (Bottom Layer) was mainly used for ground and heat dissipation. The design was carried out in the Altium Designer software environment. Large copper areas were allocated for power paths, and thermal vias were placed between the top and bottom layers to improve heat transfer. Optimal distances were maintained for free circulation of air flow between the components.

2.2. Application of design principles based on literature

In order to ensure effective thermal management during design, the methodological approaches presented in the works of Töpper et al. “Advanced Thermal Management Solutions on PCBs for High Power Applications” and Zhou et al. “Thermal Modeling and Design Optimization of PCB Vias and Pads” were taken as a basis. In this context, the use of thick copper tracks, increasing the density of vias, and optimal placement of components were applied.

2.3. Thermal simulation stage

The prepared 2-layer PCB model was subjected to thermal analysis in the Keysight Power Analyser program. The simulation was performed with the following parameters:

- Heat generation of components (W),
- PCB material: FR4, thickness 1.6 mm,
- Copper layer thickness: 70 μm ,
- Ambient temperature: 25°C,
- Convective cooling conditions: natural air flow (1.5 m/s).

Based on the simulation results, local heat accumulation was observed in some parts of the PCB and the effect of via placement was evaluated.

2.4. Experimental test

To verify the validity of the simulation results, the same PCB was fabricated in real conditions and a thermal map was taken with a thermal camera. When the motor was operated under load, the temperature values on the transistors and power paths were measured and compared with the simulation.

2.5. Analysis and optimization

As a result of the analysis, it was proposed to increase the thermal via density and place the power components more symmetrically. As a result of these adjustments, the maximum temperature on the PCB was reduced by approximately 15%, and the heat balance was more evenly distributed.

This methodology is based on real engineering approaches that can be applied to effectively optimize heat dissipation in double-sided PCB design and has been proven on both theoretical and practical grounds.

2.6 Schematic of an H-bridge circuit in Altium CAD.

H-körpü dövrəsi, sabit cərəyan (SC) mühərriklərinin istiqamət və sürət idarəetməsini təmin edən mühüm texniki qurğudur. Adətən, H-körpü dövrəsi – enerji mənbəyi, sürücü siqnalları, tranzistor açarları və mühərrikdən ibarət olan bir elektrik-elektron sistemdir. H-körpünün əsas funksiyası SC mühərrikinə tətbiq olunan gərginliyin polarizasiyasını dəyişərək onun irəli və geri fırlanmasını təmin etməkdir. PWM (Pulse Width Modulation) siqnalları və nəzarət sxemi vasitəsilə mühərrikin sürəti və fırlanma istiqaməti tənzimlənir. Şəkil 1.-də “Altium CAD”-də H körpü dövrəsinin sxemi göstərilmişdir, burada aşağıdakı işarələr verilmişdir: 1 – ilk yarım körpü sürücü dövrəsi; 2 – digər yarım körpü sürücü dövrəsi; 3 – sahə təsirli tranzistorlarla qurulmuş tam körpü motor sürücü dövrəsi; 4 – qida və motor çıxış konnektorları;

1

2



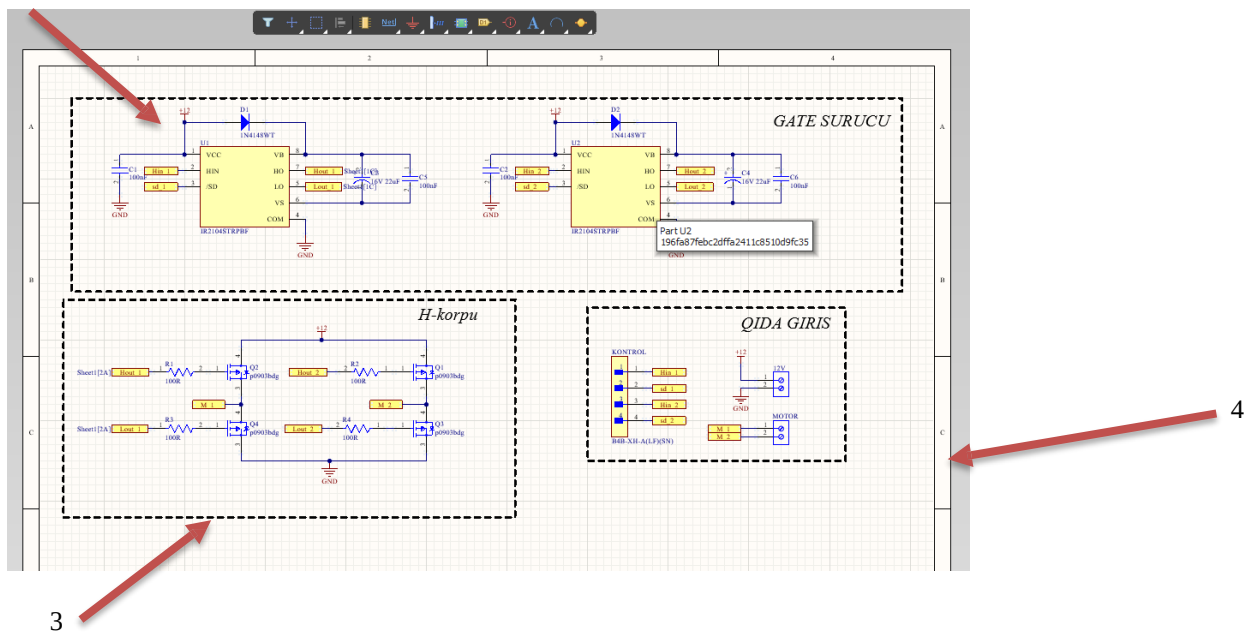


Figure 1. – Schematic of an H-bridge circuit

An H-bridge circuit is a power electronics system used to control the direction and speed of rotation of SC motors. This circuit consists mainly of four switches (MOSFETs or transistors) and allows you to control the direction and speed of the motor by changing the polarity of the voltage applied to it.

The working principle of the H-bridge is as follows:

- When S1 and S4 are open → the motor rotates forward.
- When S2 and S3 are open → the motor rotates backward.
- When all switches are closed → the motor stops.
- When S1 and S3 or S2 and S4 are open → the motor enters braking mode.

This circuit is controlled by PWM (Pulse Width Modulation) signals, allowing the motor to adjust its speed. Optimizing heat dissipation is an important consideration in the PCB design of an H-bridge circuit. The heat generated by the switching losses of transistors and the current flow in the PCB paths affects the performance and long-term durability of the motor. The main optimization methods for thermal management are:

- Wide copper traces and multilayer PCB design.
- Addition of thermal vias and use of thermal pads.
- Selection of low-resistance MOSFETs.
- Implementation of active and passive cooling solutions (heat sink and fan).

2.7 Structure of the dynamic model of heat dissipation in a PCB

The temperature change over time on a PCB is modeled by the thermal diffusion equation and the influence of heat sources generated by electrical components. The thermal diffusion equation determines the heat balance and heat flux distribution on a PCB:

$$\rho C_p \frac{\partial T}{\partial t} = k \nabla^2 T + Q$$

Here:

- ρ – PCB density of the material (kg/m³),
- C_p – PCBs heat capacity (J/ kg ·K),
- k – PCBs thermal conductivity (W/ m·K),
- T – PCBs temperature at a given point (K),
- Q – PCBs heat sources created by electronic components located (W/ m³),
- $\nabla^2 T$ – Temperature distribution within a field using the Laplace operator.

This equation shows that the temperature of a PCB depends on both the heat flux generated by the heat sources (MOSFETs, chips, etc.) and the thermal conductivity of the material. Heat is mainly dissipated in a PCB through three main mechanisms:

1. Conduction:

- o Heat is dissipated through the copper tracks and insulation layers of the PCB material.
- o Thermal conductivity depends on the thickness of the PCB and the thermal conductivity of the material.

2. Convection:

o Heat exchange occurs between the PCB surface and the environment through air flow.

o Air cooling can be passive (based on nature) or active (fan, ventilation systems).

3. Radiation:

o Heat is dissipated through infrared radiation as a result of temperature differences on the PCB surface.

o The radiation effect becomes noticeable at high temperatures.

The heat flow in a PCB is Fourier's law:

$$Q = -kA \frac{\Delta T}{\Delta x}$$

Where:

- Q – heat flux in the PCB (W),
- k – thermal conductivity of the PCB (W/m·K),
- A – heat dissipation area in the PCB (m²),
- $\Delta T/\Delta x$ – temperature gradient in the PCB (K /m).

This formula shows how heat is distributed between different points of the PCB and what role thermal resistance plays. The heat accumulation on the PCB and its change over time are modeled by the balance of heat flux generated by heat sources and heat transferred to the environment:

$$C_{\theta} \frac{dT_{pcb}(t)}{dt} = P_{loss}(t) - \frac{T_{pcb}(t) - T_{ambient}}{R_{\theta JA}}$$

Where:

- C_{θ} – PCB thermal capacity (J/K),
- $dT_{pcb}(t)/dt$ – PCB temperature rate of change over time (K/s),
- $P_{loss}(t)$ – PCB power loss over time (W),
- $R_{\theta JA}$ – PCB junction-to-ambient thermal resistance (C/W).

This equation is used to determine the temperature variation over time on the PCB and the effect of control systems on thermal distribution.

2.8 H-bridge circuit printed circuit board design and simulation of heat dissipation on the board

The designed PCB is optimized to increase the heat dissipation of electronic components and ensure long-term stability of the circuit. Based on the principles of uniform heat distribution on the PCB, expansion of high-power paths and strategic placement, the following design rules were followed: Heat sources (power transistors, driver chips) are distributed in a balanced manner to different areas of the circuit. Heat transfer connections between high-current carrying expanded and circuit layers are established. Thermal vias are placed to ensure heat dissipation between PCB layers. The width of high-current paths is increased and the electrical resistance of the circuit is reduced to prevent dangerous temperature increases.

The copper thickness of the paths at risk of overheating has been increased. Copper-filled (plated-through vias) options have been selected within the vias to increase the efficiency of heat dissipation. Gaps for air flow between the components have been maintained and appropriate placement for ventilation has been provided. When active cooling is required, integration options for additional cooling systems such as radiators and fans have been provided.

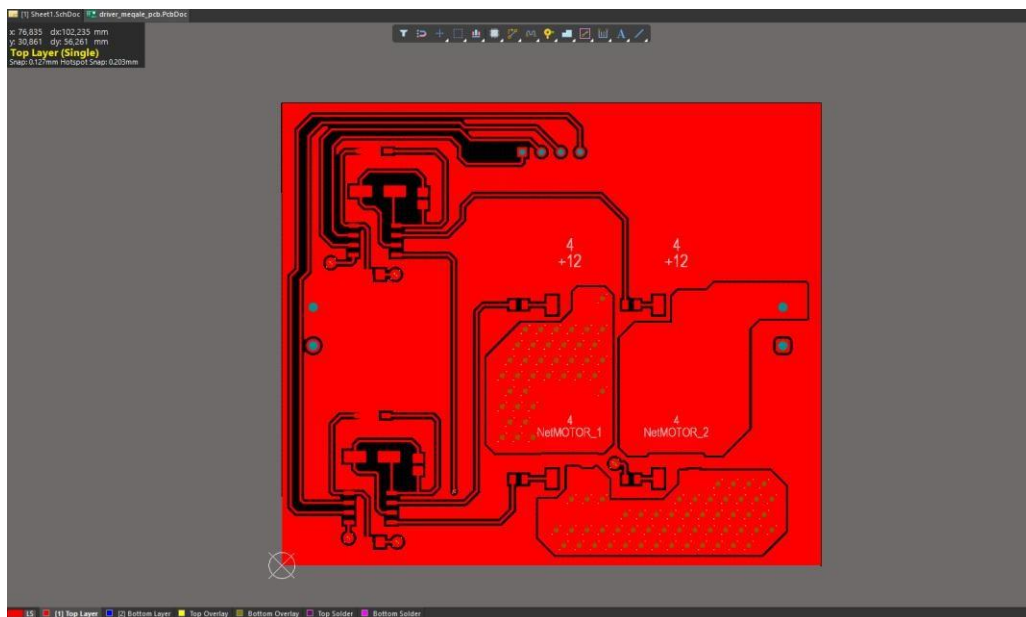


Figure 2. – PCB format of the H-bridge

The PCB design presented in Figure 2 is designed for an H-bridge based motor driver and is based on the principles of thermal management, optimization of high current paths, and efficient component placement. This design is specifically optimized to ensure the efficient operation of high-power transistors and motor control paths.

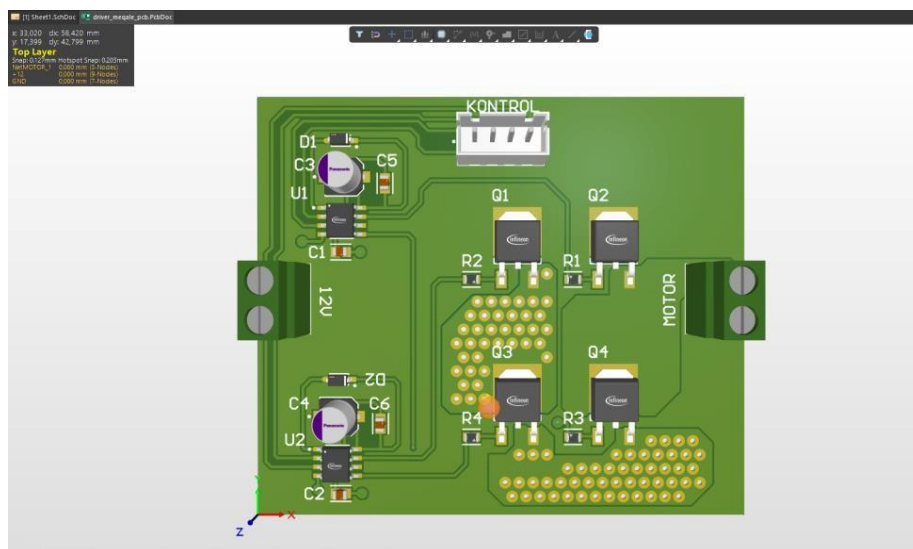


Figure 3. – 3D view of an H-bridge circuit

Figure 3. shows a 3D view of the circuit, where it can be seen that the power transistors are positioned in such a way that when they heat up, the operation of peripheral elements and their values are prevented from changing after a certain time. It is also possible to actively cool the transistors Q1, Q2, Q3 and Q4, they are symmetrically placed and will distribute the heat evenly through the cooling radiator placed on them and will be cooled by a fan.

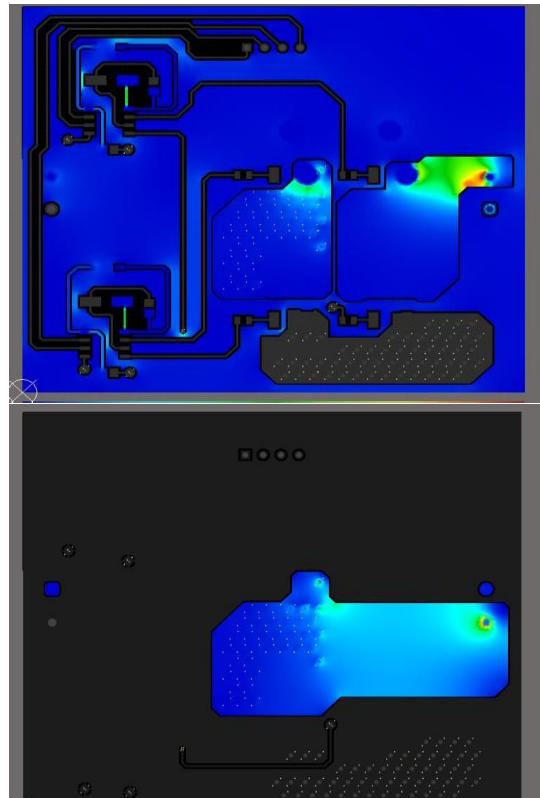


Figure 3. PCB simulation

In Figure 3, a map of the heat generated on the top and bottom layers is drawn using Keysight's power analyzer tool. For simulation, the motor parameters to be connected to the output and the current used by other circuit elements and the source current are entered. As a result, the heat generated on the card is simulated due to the current that will pass through the lines. Before simulation, configuration parameters (PCB copper density, via sizes, etc.) are entered.

3. Conclusion

The designed PCB is optimized to increase the heat dissipation of electronic components and ensure long-term stability of the circuit. Based on the principles of uniform heat distribution on the PCB, expansion of high-power paths and strategic placement, the following design rules were followed: Heat sources (power transistors, driver chips) are distributed in a balanced manner to different areas of the circuit. Heat transfer connections between high-current carrying expanded and circuit layers are established. Thermal vias are placed to ensure heat dissipation between PCB layers. The width of high-current paths is increased and the electrical resistance of the circuit is reduced to prevent dangerous temperature increases.

The copper thickness of the paths with a risk of overheating has been increased. The efficiency of heat dissipation has been increased by selecting copper-filled (plated-through vias) options within the vias. Gaps for air flow between the components have been maintained and appropriate placement for ventilation has been provided. When active cooling is required, integration possibilities for additional cooling systems such as radiators and fans have been considered.

Based on the prepared PCB design and simulation analyses, a number of improvement opportunities and future work directions have been identified.

4. References

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A photograph of a wind farm with several white wind turbines on a green, hilly landscape under a blue sky with clouds. The image is framed by a dark green curved border at the top.

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A graphic of a water splash or wave, rendered in a light green and white color scheme, positioned on the right side of the lower half of the cover.